Biology - Graduate Level Assessment 23 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. The Krebs cycle in humans occurs in the
   (a) peroxisome
   (b) cytoplasm
   (c) nucleus
   (d) intermembrane space
   (e) outer mitochondrial membrane

2. A rise in intracellular free calcium in the sea urchin oocyte causes the release of proteolytic enzymes which act to prevent polyspermy. The events just described entail the
   (a) polyspermy reaction
   (b) enzymatic reaction
   (c) proteolytic reaction
   (d) acrosomal reaction
   (e) oocyte reaction

3. A plant grows in the opposite direction of the gravitational force. This is an example of
   (a) positive chemotropism
   (b) positive phototropism
   (c) negative phototropism
   (d) negative chemotropism
   (e) positive hydrotropism

4. Sexual dimorphism is most often a result of
   (a) parallel selection.
   (b) sympatric selection.
   (c) artificial selection.
   (d) allopatric selection.
   (e) directional selection.

5. Which of the following adaptations would limit pollination by bees and promote hummingbird pollination?

(a) Flowers with a narrow opening that only small insects can access
(b) Pendant (hanging) red-colored flowers
(c) Modified petals to provide a landing space
(d) Flowers with a strong sweet fragrance
(e) Bright blue flowers that open in the early morning

True / False (10 marks)

Answer True or False

6. True or False: Does increased nerve length within the treatment volume improve trigeminal neuralgia radiosurgery?
7. True or False: Is robotically assisted laparoscopic radical prostatectomy less invasive than retropubic radical prostatectomy?
8. True or False: Is gastric electrical stimulation superior to standard pharmacologic therapy in improving GI symptoms, healthcare resources, and long-term health care benefits?
9. True or False: Juvenile osteochondritis dissecans: is it a growth disturbance of the secondary physis of the epiphysis?
10. True or False: Does binge drinking during early pregnancy increase the risk of psychomotor deficits?

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the genetic implications of red-green color blindness in the offspring of a red-green color-blind man and a woman with normal vision whose father was color blind. What are the expected phenotypic ratios?
12. Discuss the classification of algae across different kingdoms, analyzing the criteria for their placement within Fungi, Plantae, and Archaea, and the implications of these classifications.

End of Paper